DANIEL WENG

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EDUCATION

New York University, Tandon School of Engineering, New York, NY, Sep 2025 - Present
Bachelor of Science, Mechanical Engineering, Expected May 2029

BASIS Chandler, Chandler, AZ - Diploma, May 2025

EXPERIENCE

Undergraduate Research Assistant, **California Institute of Technology,** Jul 2026 - Aug 2026

- Supported the development of a robotic quadruped system designed to deploy environmental sensors in Antarctica.
- Tested mechanical mounting configurations connecting robotic arms to the quadruped, identifying fitment, stability, and integration issues.
- Conducted diagnostic testing on two robotic arm systems and communicated findings to support hardware selection and field-deployment decisions.

Engineering Intern, **Cypress Envirosystems,** Remote, Sep 2025 - Dec 2025

- Enhanced the existing housing structure for a wireless gauge reader to accommodate rotary dampers introduced in the lifting mechanism.
- Proposed design solutions to resolve hinge fitment issues and developed test plates to determine optimal dimensions for models.
- Performed tolerance stack-up analyses to predict assembly variation and define required manufacturing tolerances.

PROJECTS

Automatic Dog Feeder | Fall 2025

- Led all aspects of mechanical design for a smart pet feeder using force sensors and control algorithms to regulate portion size.
- Designed and modeled all components in Fusion 360, integrating 3D printing, laser cutting, and mechanical assembly.
- Optimized feeding mechanisms to minimize pet food waste and reduce the risk of pet obesity through precise portion calibration.

Electric Skateboard | Spring 2025

- Collaborated on the design and assembly of an electric skateboard prototype focused on improving load distribution and drivetrain efficiency.
- Proposed and developed a gear-chain drivetrain using repurposed bicycle gears and chains to transfer torque from the motor to an auxiliary axle, addressing structural load limitations.
- Assisted in mechanical assembly and woodworking, ensuring frame integrity and component alignment during prototyping.

SKILLS

CAD: Fusion 360, Autodesk Inventor, Onshape, AutoCAD
Programming and Analysis: Python, Java, Microsoft Excel
Fabrication: 3D Printing, Laser Cutting, Mechanical Assembly
Languages: English, Mandarin Chinese, Shanghainese